



BSc (Hons) in **Information Technology**
Specialized in AI

IT3012 : Intelligent Agents

Year 3 · Semester 1 · 2026 Semester 2

Faculty of Computing

Practical 01

Objective & Lecture Mapping: Recall that an intelligent agent acts within a continuous loop: Sensors generate percept sequences, and Actuators execute actions to maximize a Performance measure. Use this sheet to execute practical modifications and incrementally evaluate your design against Lecture 01 and 02 theory (from basic recall to analytical classification).

Part 1: Code Analysis & PEAS Evaluation

Task 0 : Setting up and getting the starter Code

1. Go to the [Git Hub Repo](#). Create a Fork of this Git Repo to your Personal Github Profile. Open it using your favourite IDE and follow the below instruction to complete the practical. The base code is in the "master" branch of the repo.

Task 1.1: The Environment State (`__init__`)

1. (Remember) List the four components of the PEAS framework discussed in Lecture 01.

Performance Measure
Environment
Actuators

2. (Understand) Look at the `VisualGridHuntGame.__init__` method. Which specific variables in this code represent the physical "Environment (E)" state?

```
self.width & self.height
self.walls
self.food_positions
```

3. (Analyze) Based on the variables initialized (specifically `self.opponents`), classify this baseline environment as "Single-Agent" or "Multi-Agent". Briefly justify your choice.

```
Multi-Agent
Presence of Dynamic Entities - The self.opponents variable stores a list of additional
entities ([ox, oy]) that occupy locations in the grid and execute their own movement
```

Task 1.2: The Perception Subsystem (`get_percept`)

4. (Remember) Define what a "percept sequence" is according to Lecture 01.

```
The complete history of everything the agent has ever perceived.
```

5. (Understand) Which component of the PEAS framework does the `get_percept()` method represent in our code?

```
Sensors
```

6. (Evaluate) Based on the exact dictionary returned by `get_percept()` , is this environment "Fully Observable" or "Partially Observable"? Explain why based on what the agent can and cannot see.

```
Partially Observable
What the Agent Can See
-Its own current position.
```

Task 1.3: Action Execution (`execute_action`)

7. (Understand) When `execute_action()` deducts points for hitting a wall, which PEAS component is being actively updated?

Performance Measure

8. (Evaluate) Why is it structurally important that this metric evaluates changes in the external environment state (hitting a wall) rather than the agent's internal processing effort?

It is structurally important because evaluating external environment state changes ensures the agent is measured on actual performance outcomes in the real world rather than rewarding or penalizing its internal computational effort.

Part 2: Guided Modification & Deep Theory Mapping

Follow the implementation instructions to inject a new hazard, then answer the questions to map your code changes back to the theoretical nature of the environment.

Step 2.1: Extending the Environment Initialization (`__init__`)

1. **Implementation Instructions:** Declare a new trap collection attribute (e.g., `self.toxic_traps = set()`) inside `__init__`. Populate it with coordinate tuples safely avoiding position `(0, 0)`, walls, and food.

9. (Understand) If you successfully add `self.toxic_traps` to the environment but intentionally **hide** this data from the agent's sensors, how does this specifically alter the environment's "Observability" classification?

Intentionally hiding `self.toxic_traps` from the agent's sensors reinforces the environment as Partially Observable because the agent cannot perceive the complete state of the world. Since the environment now contains physical hidden hazards that

Step 2.2: Updating the Perception Subsystem (`get_percept`)

1. **Implementation Instructions:** Locate `get_percept(self)` and add a new boolean sensor key (e.g., `'smells_toxin': tuple(self.agent_pos) in self.toxic_traps`) to the dictionary returned to the agent loop.

10. (Analyze) By adding `'smells_toxin'`, you expand the agent's percept sequence. Explain how this specific sensor helps the agent act with "Rationality" (maximizing expected utility).

The `'smells_toxin'` sensor enables rationality by allowing the agent to detect and remember penalty-causing traps, ensuring it avoids those coordinates to maximize its overall score.

Step 2.3: Modifying Action Execution & Visual Rendering (`execute_action` & `draw_grid`)

1. **Implementation Instructions:** In `execute_action`, check if the updated `self.agent_pos` intersects with `self.toxic_traps` to decrement `self.score` by 15 points. In `draw_grid`, iterate through traps to render custom purple shapes on the canvas.

11. (Remember) You programmed a severe score penalty for stepping on a trap to avoid metric exploitation. What was the classic "Vacuum World Exploit" example discussed in Lecture 01 that illustrated the danger of faulty metrics?

If the performance metric rewards the agent simply for each piece of dirt sucked up, a rational agent can exploit this by repeatedly dumping dirt back onto a clean floor and cleaning it again to indefinitely farm points without keeping the room clean.

Finalize and Lock Lab 01 Analytical Evaluation

Incomplete: 10/11 questions answered. Please provide detailed responses for all questions.

